

Coase Theorem

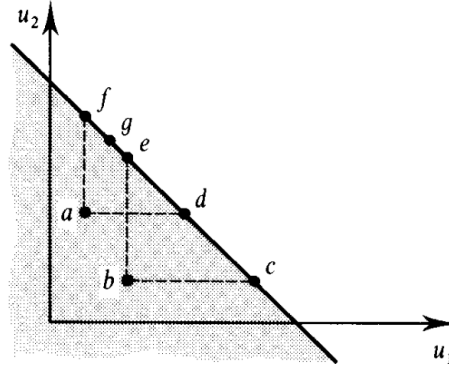
The table below shows the distribution of surplus for each combination of who owns the property rights ($R = i$ means consumer i owns the property rights) and who makes the take-it-or-leave-it offer ($B = i$ means consumer i makes the offer).

The total surplus is the same in each:

$$v_1 + v_2 = \phi_1(h^\circ) + \phi_2(h^\circ)$$

R	B	Problem	v_1	v_2
1	1	$\max_{h \geq 0, T} \phi_1(h) + T$ s.t. $\phi_2(h) - T \geq \phi_2(h^*)$	$\phi_1(h^\circ) + \phi_2(h^\circ) - \phi_2(h^*)$	$\phi_2(h^*)$
1	2	$\max_{h \geq 0, T} \phi_2(h) - T$ s.t. $\phi_1(h) + T \geq \phi_1(h^*)$	$\phi_1(h^*)$	$\phi_1(h^\circ) + \phi_2(h^\circ) - \phi_1(h^*)$
2	1	$\max_{h \geq 0, T} \phi_1(h) - T$ s.t. $\phi_2(h) + T \geq \phi_2(0)$	$\phi_1(h^\circ) + \phi_2(h^\circ) - \phi_2(0)$	$\phi_2(0)$
2	2	$\max_{h \geq 0, T} \phi_2(h) + T$ s.t. $\phi_1(h) - T \geq \phi_1(0)$	$\phi_1(0)$	$\phi_1(h^\circ) + \phi_2(h^\circ) - \phi_1(0)$

Graphically, the different outcomes can be represented as follows:



- a is where consumer 2 has the property rights and $(\phi_1(0), \phi_2(0))$. This is the outcome when no bargaining occurs and no externality occurs (because consumer 1 is forbidden to choose $h > 0$).
 - d is where consumer 1 bargains and f is where consumer 2 bargains. Whoever makes the offer extracts the full increase in the surplus.
- b is where consumer 1 has the property rights and $(\phi_1(h^*), \phi_2(h^*))$. This is the outcome when no bargaining occurs but consumer 1 chooses the competitive amount of the externality.
 - c is where consumer 1 bargains and e is where consumer 2 bargains. Whoever makes the offer extracts the full increase in the surplus.